

# Differentiable Collision Detection for a Set of Convex Primitives

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**Abstract**—Collision detection between objects is critical for simulation, control, and learning for robotic systems. However, existing collision detection routines are inherently non-differentiable, limiting their usefulness in optimization-based algorithms. In this work, we propose a fully differentiable collision-detection framework that reasons about distances between a set of composable and highly expressive convex primitive shapes. This is achieved by formulating the collision detection problem as a convex optimization problem that seeks to find the minimum uniform scaling to be applied to each object before there is an intersection. The optimization problem is fully differentiable and is able to return both the collision detection status as well as the contact points on each object.

## I. INTRODUCTION

Computing collisions has been of great interest to the computer graphics, video game, and robotics communities. Popular algorithms for collision detection are the Gilbert, Johnson, and Keerthi (GJK) algorithm [1], its updated variant enhanced-GJK [2], and Minkowski Portal Refinement (MPR) [3], [4]. These algorithms are highly efficient and robust but are inherently non-differentiable due to their logical control flow and pivoting.

This work proposes a framework for computing collision information between any pair of six convex primitive shapes; polytopes, capsules, cylinders, cones, spheres, and padded polygons as shown in Fig. 2. This is achieved by forming a convex optimization problem that solves for the minimum uniform scaling that must be applied to the objects for an intersection to exist. When objects are not in contact, the minimum scaling for an intersection is greater than one, and when there is penetration between objects, the minimum scaling is less than one. In addition to the scaling parameter detecting a collision, the contact points on each object can be calculated from this solution as well. This optimization problem is bounded, feasible, and well-defined for all configurations of the objects.

Recent advances in differentiable convex optimization allow for the sensitivities of the solution to be calculated with respect to the problem parameters. This allows for smooth derivatives of the minimum scaling as well as the contact points with respect to the configurations of the objects. The ability to differentiate through this collision detection algorithm enables the inclusion of collisions into robotic simulation, control, and learning frameworks.

This paper proceeds by describing the standard conic optimization problem and its derivatives in Section II, a general framework for collision detection in Section III, and the corresponding constraints for each of the six convex primitives shown in Fig. 2.

## II. CONIC OPTIMIZATION

This work formulates collision detection problems as generic conic problems [5]. These optimization problems have linear objectives and conic constraints of the following form:

$$\begin{aligned} & \underset{x}{\text{minimize}} && c^T x \\ & \text{subject to} && Gx + s = h, \\ & && s \in \mathcal{K}, \end{aligned} \quad (1)$$

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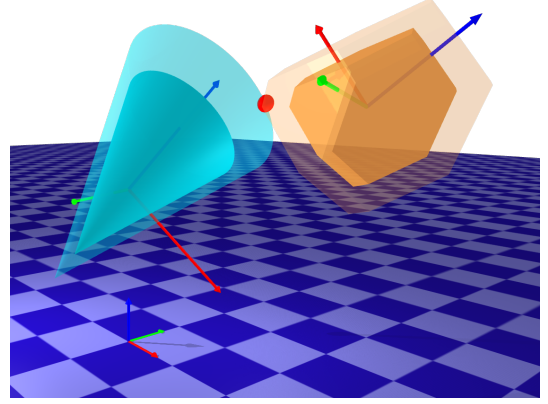


Fig. 1. Collision detection between a cone and a polytope. The algorithm works by solving an optimization problem for the minimum scaling of each object required until there is an intersection, which in this example is  $1.35 > 1$  so there is no collision. The scaled objects are translucent and the intersection between these scaled objects is shown in red.

where  $x \in \mathbb{R}^n$ ,  $c \in \mathbb{R}^n$ ,  $G \in \mathbb{R}^{m \times n}$ ,  $s \in \mathbb{R}^m$ ,  $h \in \mathbb{R}^m$ , and  $\mathcal{K} = \mathcal{K}_1 \times \dots \times \mathcal{K}_N$  is a Cartesian product of  $N$  proper convex cones. The cones required for the collision detection algorithm proposed in this work will include nonnegative orthant for linear inequalities, as well as second-order cones for convex constraints with L2 norms. The optimality conditions for problem 1 are as follows:

$$c + G^T z = 0, \quad (2)$$

$$Gx + s = h, \quad (3)$$

$$s \in \mathcal{K}, \quad (4)$$

$$z \in \mathcal{K}^*, \quad (5)$$

$$s \circ z = 0, \quad (6)$$

where a dual variable  $z \in \mathbb{R}^m$  is introduced,  $\mathcal{K}^*$  is the dual cone, and  $\circ$  is a cone product specific to each cone type [6]. These problems can be solved quickly and reliably with primal-dual interior-point methods [6], [7].

### A. Differentiating Through a Cone Program

Recent advances in differentiable convex optimization have enabled the ability to differentiate through problems of the form (1) [8]–[10]. Solutions to (1) can be differentiated with respect to the parameters  $c$ ,  $G$ , and  $h$ , in an efficient manner. This allows for the construction of algorithms that solve problem (1) while still remaining differentiable.

At the core of differentiable convex optimization is the implicit function theorem. An implicit function  $g : \mathbb{R}^a \times \mathbb{R}^b \rightarrow \mathbb{R}^a$  is defined as:

$$g(z^*, \theta) = 0. \quad (7)$$

For a solution  $z^* \in \mathbb{R}^a$ , and problem parameters  $\theta \in \mathbb{R}^b$ . Approximating (7) with a first-order Taylor series results in:

$$\frac{\partial g}{\partial z} \delta z + \frac{\partial g}{\partial \theta} \delta \theta = 0, \quad (8)$$

which can be re-arranged to solve for the sensitivities of the solution with respect to the problem parameters:

$$\frac{\partial z}{\partial \theta} = - \left( \frac{\partial g}{\partial z} \right)^{-1} \frac{\partial g}{\partial \theta}. \quad (9)$$

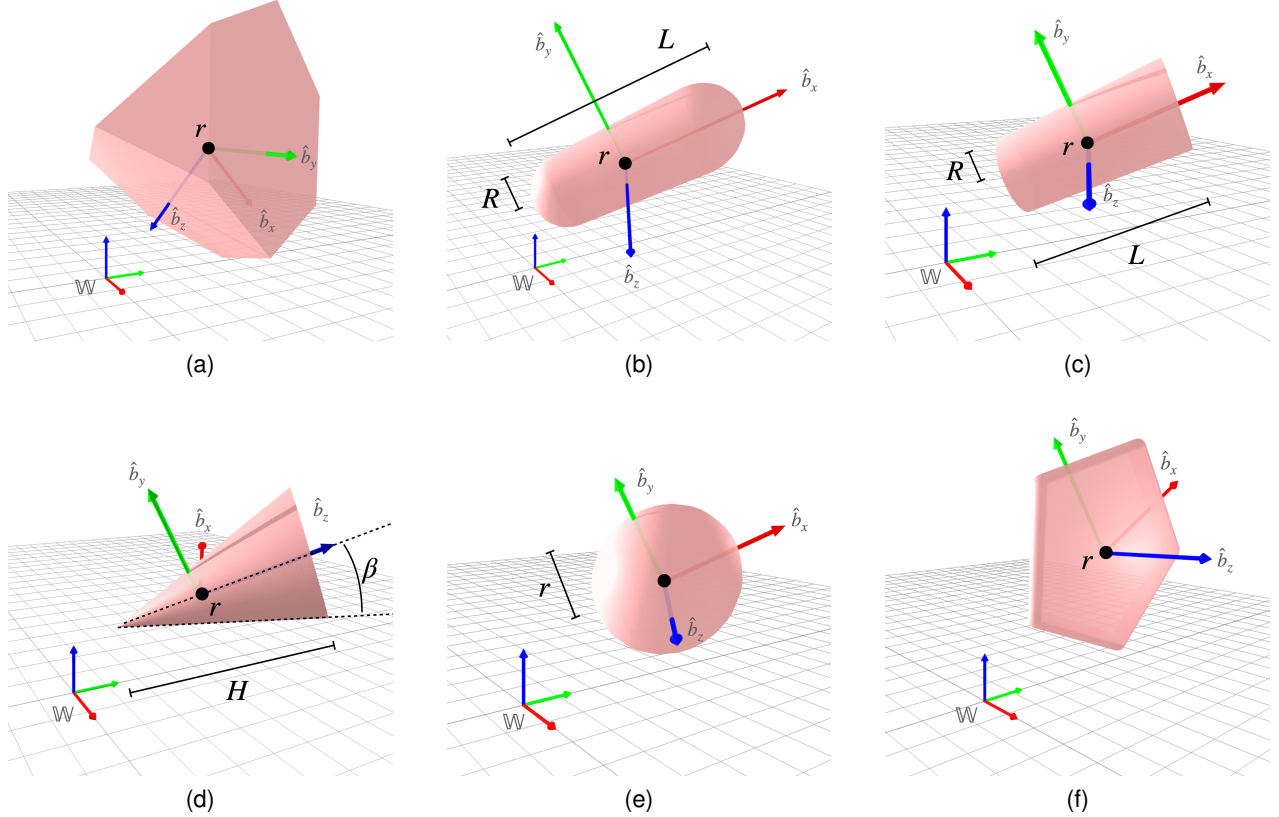


Fig. 2. Geometric descriptions of the six primitive shapes that are compatible with this differentiable collision detection algorithm. These shapes include a polytope (a), capsule (b), cylinder (c), cone (d), sphere (e), and padded polygon (f). Collision information including the collision status as well as the contact points can be computed between any of two of these primitives using a common framework.

By treating the optimality conditions of equations (2),(3), and (6), as an implicit function about the primal and dual solution, the sensitivities of the solutions with respect to the problem data can be computed. When the original optimization problem is solved using a primal-dual interior-point method as described in [6], these derivatives can be computed after the solve without any additional matrix factorizations [10]. This enables fast differentiation of conic programs that are fit for use in a differentiable collision detection algorithm.

### III. FRAMEWORK

This section details a framework for computing collision information between two convex primitives using an optimization-based approach. The core part of this framework is an optimization problem that solves for a minimum uniform scaling  $\alpha \in \mathbb{R}$  applied to both objects that result in an intersection. In the case where there is no collision between the two objects, the minimum scaling is greater than one, and when there is penetration, the minimum scaling is less than one. With the minimum scaling computed, the contact point on each object is also returned.

Each convex primitive will be described as a set  $S(\alpha)$ , which is a specific instance of the primitive scaled by some  $\alpha$ . A point  $x \in \mathbb{R}^3$  is said to be in the set  $x \in S(\alpha)$  if  $x$  is within the scaled object. This notation allows for the following formulation of the optimization

problem:

$$\begin{aligned} & \underset{x, \alpha}{\text{minimize}} && \alpha \\ & \text{subject to} && x \in \mathcal{S}_1(\alpha), \\ & && x \in \mathcal{S}_2(\alpha), \\ & && \alpha \geq 0, \end{aligned} \quad (10)$$

where the minimum scaling  $\alpha$  is computed such that  $x$  is in the interior of both objects, making  $x$  an intersection point. This optimization problem is always bounded, as well as always feasible. The boundedness comes from the constraint  $\alpha \geq 0$ , and the guarantee of feasibility comes from the fact that each object is uniformly scaled, so in the limit of  $\alpha \rightarrow \infty$ , each shape will encompass the entirety of  $\mathbb{R}^3$ , guaranteeing an intersection between objects. Another benefit to this problem formulation is that the only time the minimum scaling  $\alpha = 0$  is when the origins of the two objects are coincident, in which the problem and its derivatives are still well defined nonetheless.

#### A. Primitives

The section will detail the corresponding constraints that define the set membership for each of the six scaled primitives. Each object is defined with an attached reference frame  $\mathbb{B}$  with an origin  $r \in \mathbb{R}^3$  expressed in a world frame  $\mathbb{W}$ . The uniform scaling of these objects is always centered about this position  $r$ , and when the scaling parameter  $\alpha = 0$ , the object is simply a point at  $r$ . The configuration of an object is defined by a rotation matrix  ${}^{\mathbb{W}}Q^{\mathbb{B}} \in \mathbb{R}^{3 \times 3}$  relating the world frame to the object-fixed frame, and the position of the origin of this object-fixed frame,  $r$ .

1) *Polytopes*: A polytope is a convex shape in  $\mathbb{R}^3$  defined by a set of halfspace constraints, an example of which is shown in Fig. 2a. This polytope has an attached reference frame  $\mathbb{B}$  with an origin at  $r$  and is described as the set of points  $y \in \mathbb{R}^3$  such that  $Ay \leq b$  for  $y$  expressed in  $\mathbb{B}$ , where  $A \in \mathbb{R}^{m,3}$  and  $b \in \mathbb{R}^m$  represent the  $m$  halfspace constraints. This polytope can be scaled by  $\alpha$ , resulting in the following constraint for  $x$  to be inside the polytope:

$$A^{\mathbb{B}}Q^{\mathbb{W}}(x-r) \leq \alpha b, \quad (11)$$

which is a linear inequality and therefore convex. The scaling parameter  $\alpha$  scales the vector  $b$ , resulting in uniform scaling of all halfspace constraints and subsequent uniform scaling of the polytope.

2) *Capsules*: A capsule can be defined by the set of points within some radius  $R$  of a line segment, as shown in Fig. 2b. This internal line segment is fixed to the  $x$  axis of the attached reference frame  $\mathbb{B}$ , and the end points of this line segment are some distance  $L$  apart. The scaled constraints for this primitive are that the point  $x$  must be within a scaled radius of the line segment, where the distance of the endpoints of the line segment from  $r$  is also scaled:

$$\|x - (r + \gamma \hat{b}_x)\| \leq \alpha R, \quad (12)$$

$$-\alpha \frac{L}{2} \leq \gamma \leq \alpha \frac{L}{2}, \quad (13)$$

where  $\hat{b}_x = {}^{\mathbb{W}}Q^{\mathbb{B}}[1, 0, 0]^T$ . These constraints contain one second-order cone constraint and a linear inequality.

3) *Cylinder*: The description of a cylinder is shown in Fig. 2c, with an orientation, a radius  $R$ , and a length  $L$ . The constraints for this primitive are the same as for the capsule in equations (12) and (13), with the introduction of two new scaled halfspace constraints that give the cylinder its flat ends:

$$[x - (r - \alpha \frac{L}{2} \hat{b}_x)]^T \hat{b}_x \geq 0, \quad (14)$$

$$[x - (r + \alpha \frac{L}{2} \hat{b}_x)]^T \hat{b}_x \leq 0, \quad (15)$$

where again  $\hat{b}_x = {}^{\mathbb{W}}Q^{\mathbb{B}}[1, 0, 0]^T$ .

4) *Cones*: As shown in Fig. 2d, a cone can be described with a height  $H$ , and a half angle  $\beta$ . The origin of the object-fixed frame  $r$  is at the halfway point between the pointed end of the cone and the flat face, and  $\alpha$  scales the distance of these two ends from the center point  $r$ :

$$c = r - \alpha \frac{H}{2} \hat{b}_z, \quad (16)$$

$$d = r + \alpha \frac{H}{2} \hat{b}_z, \quad (17)$$

$$\tilde{x} = {}^{\mathbb{B}}Q^{\mathbb{W}}(x - c), \quad (18)$$

$$\|\tilde{x}_{1:2}\| \leq \tan(\beta) \tilde{x}_3, \quad (19)$$

$$(x - d)^T \hat{b}_z \leq 0, \quad (20)$$

where  $\hat{b}_z = {}^{\mathbb{W}}Q^{\mathbb{B}}[0, 0, 1]^T$ .

5) *Spheres*: Spheres are the simplest primitive, shown in Fig. 2e, where  $x$  must be within some scaled radius of the center  $r$ :

$$\|x - r\| \leq \alpha R. \quad (21)$$

A point could also be represented as a sphere in the limit as  $R \rightarrow \infty$ .

6) *Padded Polygon*: A “padded” polygon is defined as the set of points within some radius  $R$  of a two-dimensional polygon. Shown in Fig. 2f, the first two basis vectors of  $\mathbb{B}$  span the polygon, and the polygon itself is defined with two-dimensional halfspace constraints

$Ay \leq \alpha b$ . This polygon is scaled in the same fashion as the polytope, and results in the following constraints:

$$\|x - (r + {}^{\mathbb{W}}\tilde{Q}^{\mathbb{B}}y)\| \leq \alpha R, \quad (22)$$

$$Ay \leq \alpha b, \quad (23)$$

where  ${}^{\mathbb{W}}\tilde{Q}^{\mathbb{B}} \in \mathbb{R}^{3 \times 2}$  is the first two columns of  ${}^{\mathbb{W}}Q^{\mathbb{B}}$ .

## B. Contact Point

The point on the two scaled primitives that intersect is referred to as  $x^*$ , but unless  $\alpha^* = 1$ , this point does not exist on the surface of the primitives. The corresponding contact point for each primitive,  $p \in \mathbb{R}^3$ , is calculated using the optimal  $x^*$  and  $\alpha^*$  from (10):

$$p = r + \frac{x^* - r}{\alpha}. \quad (24)$$

This operation is differentiable and allows for the calculation of the sensitivities of the contact points to the configurations of the primitives.

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